Lab Manual: Introduction to Memory Forensics with Volatility 3

1. Overview: The Volatility Framework

Volatility is a highly powerful memory forensics tool designed to extract information from memory images (dumps) of Windows, macOS, and Linux systems[1]. It is an essential component of a forensic toolkit because it allows investigators to analyze evidence that is never written to the hard disk.

By using Volatility 3, analysts can perform critical tasks such as:

- Listing running processes.
- Checking active network connections.
- Extracting files and analyzing the Windows Registry.
- Utilizing third-party plugins developed by a large community.

Installation of Volatility Framework 3

System Requirements

- OS: Windows / Linux / macOS
- Python version: **Python 3.8 – 3.11 (recommended: 3.10)**
- Memory image: `.raw`, `.mem`, `.dd`, `.lime`, `.aff4`

Step 1: Install Python

Verify Python installation:

```
python --version
```

If not installed, download Python from the official website and ensure:

- ✔ “Add Python to PATH” is checked

Step 2: Download Volatility 3

```
git clone https://github.com/volatilityfoundation/volatility3.git
cd volatility3
```

Or download ZIP and extract it.

Step 3: Verify Installation

```
python vol.py -h
```

If the help menu appears, Volatility 3 is installed successfully.

2. Lab Exercises: Volatility 3 Commands

Prerequisite: Ensure you are running Volatility 3 in Windows PowerShell or your preferred terminal.

Step 1: Verification

Check the installed version of Volatility to ensure the tool is running correctly.

Command:

PowerShell

```
python vol.py -v
```

```
(frida) E:\Volitility 3\volatility3>python vol.py -v
INFO     volatility3.cli: Volatility plugins path: ['E:\\Volitility 3\\volatili
ework\\plugins']
INFO     volatility3.cli: Volatility symbols path: ['E:\\Volitility 3\\volatili
ework\\symbols']
Volatility 3 Framework 2.27.1
usage: vol.py [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
              [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
              [-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
              [--save-config SAVE_CONFIG] [--clear-cache]
              [--cache-path CACHE_PATH] [--offline | -u URL]
              [--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]]
              [-r RENDERER] [--single-location SINGLE_LOCATION]
              [--stackers [STACKERS ...]]
              [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
              PLUGIN ...
vol.py: error: Please select a plugin to run (see 'vol.py --help' for options

(frida) E:\Volitility 3\volatility3>
```

Step 2: Image Identification

Obtain basic information about the memory image you are analyzing.

Command:

PowerShell

```
python vol.py -f [ImageName] windows.info
```

```
Variable        Value

Kernel Base     0xf8043cc00000
DTB     0x1aa000
Symbols file:///E:/Volitility%203/volatility3/volatility3/symbols/windows/nt
krnlmp.pdb/769C521E4833ECF72E21F02BF33691A5-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 FileLayer
KdVersionBlock  0xf8043d80f368
Major/Minor     15.19041
MachineType     34404
KeNumberProcessors      4
SystemTime      2021-04-30 17:52:19+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Tue Oct 11 07:04:26 1977

(frida) E:\Volitility 3\volatility3>
```

Step 3: Process Analysis

View the list of processes running at the time of the memory dump.

Command:

PowerShell

```
python vol.py -f [ImageName] windows.pslist | more
```

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.pslist | more
Progress:  0.00         Scanning layer_name using PdbSignatureScanneProgress:  0.00         Scanning layer_name using PdbSignatureScanneProg
ress:  100.00           PDB scanning finished                   Volatility 3 Framework 2.27.1

PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0xbf0f64a63080  132     -       N/A     False   2021-04-30 12:39:40.000000 UTC  N/A     Disabled
108     4       Registry        0xbf0f64bc6040  4       -       N/A     False   2021-04-30 12:39:38.000000 UTC  N/A     Disabled
396     4       smss.exe        0xbf0f66967040  2       -       N/A     False   2021-04-30 12:39:40.000000 UTC  N/A     Disabled
492     484     csrss.exe       0xbf0f6adb6080  13      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
568     484     wininit.exe     0xbf0f6b67a080  1       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
584     560     csrss.exe       0xbf0f6b681080  16      -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
668     560     winlogon.exe    0xbf0f6b6db080  5       -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
712     568     services.exe    0xbf0f6b6da080  11      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
736     568     lsass.exe       0xbf0f6b6fb0c0  10      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
856     712     svchost.exe     0xbf0f6b7042c0  20      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
884     568     fontdrvhost.ex  0xbf0f6b70b1c0  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
892     668     fontdrvhost.ex  0xbf0f6b7091c0  5       -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
976     712     svchost.exe     0xbf0f6b785340  12      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
320     712     svchost.exe     0xbf0f6b78a2c0  6       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
564     668     LogonUI.exe     0xbf0f6b7b7100  0       -       1       False   2021-04-30 12:39:44.000000 UTC  2021-04-30 17:39:58.000000 UTC  Disabled560
    668     dwm.exe 0xbf0f6b7ba080  21      -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1080    712     svchost.exe     0xbf0f6be74340  2       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1088    712     svchost.exe     0xbf0f6be77080  3       -       0       Fals-- More  --
```

Step 4: Filtering Processes

Filter the process list to search for a specific keyword (e.g., "chrome").

Command:

PowerShell

```
python vol.py -f [ImageName] windows.pslist | Select-String chrome
or
python vol.py -f [ImageName] windows.pslist | findstr "chrome"
```

11

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.pslist | findstr "chrome"
1328ress435200.0chrome.exe      0xbf0f6d53e080in26hed    -       1       False   2021-04-30 17:44:52.000000 UTC  N/A     Disabled
6764    1328    chrome.exe      0xbf0f6d748080  7       -       1       False   2021-04-30 17:44:52.000000 UTC  N/A     Disabled
4508    1328    chrome.exe      0xbf0f6be38340  10      -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
1840    1328    chrome.exe      0xbf0f6cbc5080  12      -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
5492    1328    chrome.exe      0xbf0f6a897080  7       -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
2432    1328    chrome.exe      0xbf0f6a9e7080  19      -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
5888    1328    chrome.exe      0xbf0f6d2ea340  13      -       1       False   2021-04-30 17:44:57.000000 UTC  N/A     Disabled
460     1328    chrome.exe      0xbf0f6d591300  12      -       1       False   2021-04-30 17:48:07.000000 UTC  N/A     Disabled
4108    1328    chrome.exe      0xbf0f667f1300  12      -       1       False   2021-04-30 17:48:07.000000 UTC  N/A     Disabled
3380    1328    chrome.exe      0xbf0f6d182080  6       -       1       False   2021-04-30 17:48:07.000000 UTC  N/A     Disabled
```

Step 5: Handle Analysis

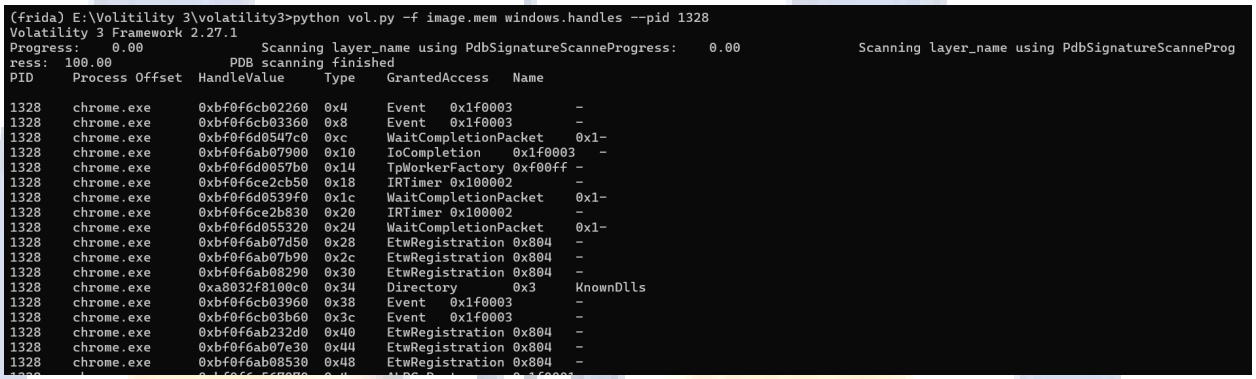
Find all handles opened by a specific Process ID (PID). *Note: Replace 1328 with your target PID.*

Command:

PowerShell

```
python vol.py -f [ImageName] windows.handles --pid 1328
```

12



```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.handles --pid 1328
Volatility 3 Framework 2.27.1
Progress:    0.00            Scanning layer_name using PdbSignatureScanneProgress:    0.00            Scanning layer_name using PdbSignatureScanneProg
ress:  100.00               PDB scanning finished
PID     Process Offset  HandleValue     Type    GrantedAccess   Name

1328    chrome.exe      0xbf0f6cb02260  0x4     Event   0x1f0003        -
1328    chrome.exe      0xbf0f6cb03360  0x8     Event   0x1f0003        -
1328    chrome.exe      0xbf0f6d0547c0  0xc     WaitCompletionPacket    0x1-
1328    chrome.exe      0xbf0f6ab07900  0x10    IoCompletion    0x1f0003    -
1328    chrome.exe      0xbf0f6d0057b0  0x14    TpWorkerFactory 0xf00ff -
1328    chrome.exe      0xbf0f6ce2cb50  0x18    IRTimer 0x100002        -
1328    chrome.exe      0xbf0f6d0539f0  0x1c    WaitCompletionPacket    0x1-
1328    chrome.exe      0xbf0f6ce2b830  0x20    IRTimer 0x100002        -
1328    chrome.exe      0xbf0f6d055320  0x24    WaitCompletionPacket    0x1-
1328    chrome.exe      0xbf0f6ab07d50  0x28    EtwRegistration 0x804   -
1328    chrome.exe      0xbf0f6ab07b90  0x2c    EtwRegistration 0x804   -
1328    chrome.exe      0xbf0f6ab08290  0x30    EtwRegistration 0x804   -
1328    chrome.exe      0xa8032f8100c0  0x34    Directory       0x3     KnownDlls
1328    chrome.exe      0xbf0f6cb03960  0x38    Event   0x1f0003        -
1328    chrome.exe      0xbf0f6cb03b60  0x3c    Event   0x1f0003        -
1328    chrome.exe      0xbf0f6ab232d0  0x40    EtwRegistration 0x804   -
1328    chrome.exe      0xbf0f6ab07e30  0x44    EtwRegistration 0x804   -
1328    chrome.exe      0xbf0f6ab08530  0x48    EtwRegistration 0x804   -
```

Step 6: Filtering File Handles

Find handles specifically related to files for a target PID.

Command:

PowerShell

```
python vol.py -f [ImageName] windows.handles --pid 1328 | Select-String File | more
Or
python vol.py -f [ImageName] windows.handles --pid 1328 | findstr "File" | more
```

13

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.handles --pid 1328 | findstr "File" | more
1328    chrome.exe      0xbf0f6ac8c9f0  0x70    File    0x100020        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93
1328    chrome.exe      0xbf0f6ac8a470  0x8c    File    0x100001        \Device\KsecDD
1328    chrome.exe      0xbf0f6ac8bd70  0x9c    File    0x100001        \Device\CNG
1328    chrome.exe      0xbf0f6abe2080  0x294   File    0x120089        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93\icudtl.
dat
1328    chrome.exe      0xbf0f6abe2850  0x29c   File    0x120089        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93\v8_cont
ext_snapshot.bin
1328    chrome.exe      0xbf0f6abe3660  0x33c   File    0x12019f        \Device\HarddiskVolume2\Users\John Doe\AppData\Local\Google\Chrome\User Data\Browser
Metrics\BrowserMetrics-608C4214-530.pma
1328    chrome.exe      0xbf0f6abe3b10  0x344   File    0x120089        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93\chrome_
100_percent.pak
1328    chrome.exe      0xbf0f6abe3e30  0x34c   File    0x120089        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93\chrome_
200_percent.pak
1328    chrome.exe      0xbf0f6abe4150  0x354   File    0x120089        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93\Locales
\en-US.pak
1328    chrome.exe      0xbf0f6abc6970  0x35c   File    0x120089        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93\resourc
es.pak
1328    chrome.exe      0xbf0f6abe4600  0x444   File    0x100020        \Device\HarddiskVolume2\Windows\WinSxS\amd64_microsoft.windows.common-controls_6595b
64144ccf1df_6.0.19041.844_none_ca00b6081b84eb1d
1328    chrome.exe      0xbf0f6c5bb2a0  0x474   File    0x100080        \Device\Nsi
1328    chrome.exe      0xbf0f6db9c240  0x490   File    0x100020        \Device\HarddiskVolume2\Windows\WinSxS\amd64_microsoft.windows.common-controls_6595b
64144ccf1df_6.0.19041.844_none_ca00b6081b84eb1d
1328    chrome.exe      0xbf0f6ac8b280  0x5a4   File    0x100020        \Device\HarddiskVolume2\Program Files\Google\Chrome\Application\90.0.4430.93
1328    chrome.exe      0xbf0f6db9d1e0  0x5e4   File    0x12019f        \Device\HarddiskVolume2\Users\John Doe\AppData\Local\Microsoft\Windows\Explorer\icon
cache_32.db
1328    chrome.exe      0xbf0f6db935a0  0x628   File    0x100001        \Device\HarddiskVolume2\Windows\System32\drivers\etc
1328    chrome.exe      0xbf0f6abe2d00  0x698   File    0x120089        \Device\DeviceApi\CMApi
```

Step 7: Dumping Specific Files

Dump a file from a process using its virtual address.

Command:

Create a Directory named “dump” then execute the command

PowerShell

```
python vol.py -f [ImageName] -o "dump" windows.dumpfile --pid 1328 --virtaddr
0xbf0f6abe9740
Or
python vol.py -f image.mem -o "dump" windows.dumpfiles --pid 1328 --virtaddr
0xbf0f6abe9740
```

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem -o "dump" windows.dumpfiles --pid 1328 --virtaddr 0xbf0f6abe9740
Volatility 3 Framework 2.27.1
Progress:  100.00               PDB scanning finished
Cache   FileObject      FileName        Result

DataSectionObject       0xbf0f6abe9740  History Error dumping file
SharedCacheMap  0xbf0f6abe9740  History file.0xbf0f6abe9740.0xbf0f6c107c70.SharedCacheMap.History.vacb
```

Step 8: Bulk File Dump

Dump all files associated with a specific Process ID (e.g., PID 2520).

Command:

PowerShell

`python vol.py -f [ImageName] windows.dumpfiles.DumpFiles --pid 2520`

15

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.dumpfiles.DumpFiles --pid 2520
Volatility 3 Framework 2.27.1
Progress:  100.00               PDB scanning finished
Cache   FileObject      FileName        Result

DataSectionObject       0xbf0f6abdf1a0  notepad.exe.mui file.0xbf0f6abdf1a0.0xbf0f6cdaa3d0.DataSectionObject.notepad.exe.mui.dat
DataSectionObject       0xbf0f6abe0dc0  StaticCache.dat Error dumping file
SharedCacheMap  0xbf0f6abe0dc0  StaticCache.dat file.0xbf0f6abe0dc0.0xbf0f6ca7cc70.SharedCacheMap.StaticCache.dat.vacb
DataSectionObject       0xbf0f6d7c53b0  notepad.exe.mui file.0xbf0f6d7c53b0.0xbf0f6cdaa3d0.DataSectionObject.notepad.exe.mui.dat
DataSectionObject       0xbf0f6cb9dd80  StaticCache.dat Error dumping file
SharedCacheMap  0xbf0f6cb9dd80  StaticCache.dat file.0xbf0f6cb9dd80.0xbf0f6ca7cc70.SharedCacheMap.StaticCache.dat.vacb
ImageSectionObject      0xbf0f6a877700  bcryptprimitives.dll    file.0xbf0f6a877700.0xbf0f6a397010.ImageSectionObject.bcryptprimitives.dll.img
ImageSectionObject      0xbf0f6b7e9bd0  uxtheme.dll     file.0xbf0f6b7e9bd0.0xbf0f6b79dcd0.ImageSectionObject.uxtheme.dll.img
ImageSectionObject      0xbf0f6c6a4530  MrmCoreR.dll    file.0xbf0f6c6a4530.0xbf0f6c3259a0.ImageSectionObject.MrmCoreR.dll.img
ImageSectionObject      0xbf0f6d7c61c0  notepad.exe     file.0xbf0f6d7c61c0.0xbf0f6ca7ec70.ImageSectionObject.notepad.exe.img
ImageSectionObject      0xbf0f6cb9be40  oleacc.dll      file.0xbf0f6cb9be40.0xbf0f6c483c50.ImageSectionObject.oleacc.dll.img
ImageSectionObject      0xbf0f6c4d0cd0  efswrt.dll      file.0xbf0f6c4d0cd0.0xbf0f6c9c5c80.ImageSectionObject.efswrt.dll.img
ImageSectionObject      0xbf0f6c4d5c80  TextShaping.dll file.0xbf0f6c4d5c80.0xbf0f6ca2b240.ImageSectionObject.TextShaping.dll.img
ImageSectionObject      0xbf0f6bfb5380  mpr.dll file.0xbf0f6bfb5380.0xbf0f6adaac70.ImageSectionObject.mpr.dll.img
ImageSectionObject      0xbf0f6c3a4c60  comctl32.dll    file.0xbf0f6c3a4c60.0xbf0f6c39aa20.ImageSectionObject.comctl32.dll.img
ImageSectionObject      0xbf0f6c295810  urlmon.dll      file.0xbf0f6c295810.0xbf0f6c2b0c70.ImageSectionObject.urlmon.dll.img
ImageSectionObject      0xbf0f6c3a5d90  iertutil.dll    file.0xbf0f6c3a5d90.0xbf0f6c32ad20.ImageSectionObject.iertutil.dll.img
ImageSectionObject      0xbf0f6c3addb0  TextInputFramework.dll  file.0xbf0f6c3addb0.0xbf0f6c48eb40.ImageSectionObject.TextInputFramework.dll.img
ImageSectionObject      0xbf0f6b7ed280  WinTypes.dll    file.0xbf0f6b7ed280.0xbf0f6be1a2e0.ImageSectionObject.WinTypes.dll.img
ImageSectionObject      0xbf0f6bea6ef0  twinapi.appcore.dll     file.0xbf0f6bea6ef0.0xbf0f6be34a20.ImageSectionObject.twinapi.appcore.dll.img
ImageSectionObject      0xbf0f6bfab420  policymanager.dll       file.0xbf0f6bfab420.0xbf0f6bf5fa20.ImageSectionObject.policymanager.dll.img
ImageSectionObject      0xbf0f6b7ee6d0  propsys.dll     file.0xbf0f6b7ee6d0.0xbf0f6be36a20.ImageSectionObject.propsys.dll.img
ImageSectionObject      0xbf0f6b7e9590  CoreUIComponents.dll    file.0xbf0f6b7e9590.0xbf0f6be19b20.ImageSectionObject.CoreUIComponents.dll.img
ImageSectionObject      0xbf0f6b7ea3a0  CoreMessaging.dll       file.0xbf0f6b7ea3a0.0xbf0f6be17d00.ImageSectionObject.CoreMessaging.dll.img
ImageSectionObject      0xbf0f6a8786a0  gdi32full.dll   file.0xbf0f6a8786a0.0xbf0f6acc9250.ImageSectionObject.gdi32full.dll.img
ImageSectionObject      0xbf0f6654fd50  msvcp110_win.dll        file.0xbf0f6654fd50.0xbf0f6b63bd90.ImageSectionObject.msvcp110_win.dll.img
ImageSectionObject      0xbf0f664b5410  windows.storage.dll     file.0xbf0f664b5410.0xbf0f6b6f8cc0.ImageSectionObject.windows.storage.dll.img
ImageSectionObject      0xbf0f6b7299f0  kernel.appcore.dll      file.0xbf0f6b7299f0.0xbf0f6b707d30.ImageSectionObject.kernel.appcore.dll.img
ImageSectionObject      0xbf0f664b10e0  ntmarta.dll     file.0xbf0f664b10e0.0xbf0f6b6b4a20.ImageSectionObject.ntmarta.dll.img
ImageSectionObject      0xbf0f6aef7bc0  wldp.dll        file.0xbf0f6aef7bc0.0xbf0f6b668c50.ImageSectionObject.wldp.dll.img
```

Step 9: Command Line History

View the executed programs and their command-line arguments.

Command:

PowerShell

`python vol.py -f [ImageName] windows.cmdline.CmdLine`

16

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.cmdline.CmdLine
Volatility 3 Framework 2.27.1
Progress:  100.00               PDB scanning finished
PID     Process Args

4       System  -
108     Registry        -
396     smss.exe        \SystemRoot\System32\smss.exe
492     csrss.exe       %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=bas
esrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
568     wininit.exe     wininit.exe
584     csrss.exe       %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=bas
esrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
668     winlogon.exe    winlogon.exe
712     services.exe    C:\Windows\system32\services.exe
736     lsass.exe       C:\Windows\system32\lsass.exe
856     svchost.exe     C:\Windows\system32\svchost.exe -k DcomLaunch -p
884     fontdrvhost.ex  "fontdrvhost.exe"
892     fontdrvhost.ex  "fontdrvhost.exe"
976     svchost.exe     C:\Windows\system32\svchost.exe -k RPCSS -p
320     svchost.exe     C:\Windows\system32\svchost.exe -k DcomLaunch -p -s LSM
564     LogonUI.exe     -
560     dwm.exe "dwm.exe"
1080    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNoNetwork -p
1088    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s lmhosts
1164    svchost.exe     C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService
1172    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s TimeBrokerSvc
1204    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s DispBrokerDesktopSvc
1296    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
1336    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s EventLog
1376    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s ProfSvc
1440    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s nsi
1452    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s UserManager
1556    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp
1660    VBoxService.ex  C:\Windows\System32\VBoxService.exe
1684    svchost.exe     C:\Windows\System32\svchost.exe -k NetworkService -p -s NlaSvc
1784    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s EventSystem
1804    svchost.exe     C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain
1824    svchost.exe     C:\Windows\System32\svchost.exe -k netsvcs -p -s Themes
1880    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p
```

Step 10: Network Analysis

See active network connections and listening programs.

Command:

PowerShell

python vol.py -f [ImageName] windows.netstat

```
(frida) E:\Volitility 3\volatility3>python vol.py -f image.mem windows.netstat
Volatility 3 Framework 2.27.1
Progress:  100.00               PDB scanning finished
Offset  Proto   LocalAddr       LocalPort       ForeignAddr     ForeignPort     State   PID     Owner   Created

0xbf0f6a535aa0  TCPv4   10.0.2.15       49846   96.90.32.107    7680    SYN_SENT        2116    svchost.exe     2021-04-30 17:52:01.000000 UTC
0xbf0f6d8a1010  TCPv4   10.0.2.15       49771   185.70.41.35    443     CLOSE_WAIT      1840    chrome.exe      2021-04-30 17:44:57.000000 UTC
0xbf0f6cbb9530  TCPv4   10.0.2.15       49772   185.70.41.35    443     FIN_WAIT2       1840    chrome.exe      2021-04-30 17:44:58.000000 UTC
0xbf0f6d0c64a0  TCPv4   10.0.2.15       49843   204.79.197.222  443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:26.000000 UTC
0xbf0f6ad1fad0  TCPv4   10.0.2.15       49847   52.230.222.68   443     ESTABLISHED     2812    svchost.exe     2021-04-30 17:52:17.000000 UTC
0xbf0f6ca71a20  TCPv4   10.0.2.15       49769   142.250.190.14  443     CLOSE_WAIT      1840    chrome.exe      2021-04-30 17:44:55.000000 UTC
0xbf0f6cfd17f0  TCPv4   10.0.2.15       49777   35.186.220.63   443     CLOSE_WAIT      1840    chrome.exe      2021-04-30 17:44:58.000000 UTC
0xbf0f6ad16050  TCPv4   10.0.2.15       49829   142.250.191.208 443     ESTABLISHED     5624    svchost.exe     2021-04-30 17:49:58.000000 UTC
0xbf0f6c85bb20  TCPv4   10.0.2.15       49775   185.70.41.35    443     FIN_WAIT2       1840    chrome.exe      2021-04-30 17:44:58.000000 UTC
0xbf0f6d51c4a0  TCPv4   10.0.2.15       49838   13.107.3.254    443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:23.000000 UTC
0xbf0f6d5c8a70  TCPv4   10.0.2.15       49797   172.217.4.74    443     CLOSE_WAIT      1840    chrome.exe      2021-04-30 17:48:27.000000 UTC
0xbf0f6d51c010  TCPv4   10.0.2.15       49763   172.217.4.35    443     CLOSE_WAIT      1840    chrome.exe      2021-04-30 17:44:53.000000 UTC
0xbf0f6c6352b0  TCPv4   10.0.2.15       49842   52.113.196.254  443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:25.000000 UTC
0xbf0f6d525a20  TCPv4   10.0.2.15       49845   23.101.202.202  443     ESTABLISHED     1156    MsMpEng.exe     2021-04-30 17:51:36.000000 UTC
0xbf0f6a896ae0  TCPv4   10.0.2.15       49773   185.70.41.35    443     FIN_WAIT2       1840    chrome.exe      2021-04-30 17:44:58.000000 UTC
0xbf0f6d5d1ac0  TCPv4   10.0.2.15       49770   185.70.41.35    443     FIN_WAIT2       1840    chrome.exe      2021-04-30 17:44:57.000000 UTC
0xbf0f6cd4fa20  TCPv4   10.0.2.15       49837   204.79.197.200  443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:18.000000 UTC
0xbf0f6c7104d0  TCPv4   10.0.2.15       49778   185.70.41.130   443     ESTABLISHED     1840    chrome.exe      2021-04-30 17:45:00.000000 UTC
0xbf0f66443b40  TCPv4   10.0.2.15       49839   23.35.68.233    443     CLOSE_WAIT      5104    SearchApp.exe   2021-04-30 17:51:24.000000 UTC
0xbf0f6a53ca20  TCPv4   10.0.2.15       49833   52.230.222.68   443     ESTABLISHED     2812    svchost.exe     2021-04-30 17:50:07.000000 UTC
0xbf0f66ff7910  TCPv4   0.0.0.0 135     0.0.0.0 0       LISTENING       976     svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff7910  TCPv6   ::      135     ::      0       LISTENING       976     svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff6e10  TCPv4   0.0.0.0 135     0.0.0.0 0       LISTENING       976     svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f6a5a7230  TCPv4   10.0.2.15       139     0.0.0.0 0       LISTENING       4       System  2021-04-30 12:39:44.000000 UTC
0xbf0f6a5a6e10  TCPv4   0.0.0.0 445     0.0.0.0 0       LISTENING       4       System  2021-04-30 17:39:47.000000 UTC
0xbf0f6a5a6e10  TCPv6   ::      445     ::      0       LISTENING       4       System  2021-04-30 17:39:47.000000 UTC
0xbf0f6a837e10  TCPv4   0.0.0.0 5040    0.0.0.0 0       LISTENING       4128    svchost.exe     2021-04-30 17:39:48.000000 UTC
```

Step 11: Credential Harvesting

Dump Windows user password hashes.

Command:

PowerShell

python vol.py -f [ImageName] windows.hashdump.Hashdump

```
Volatility 3 Framework 2.27.1
E:\Volitility 3\volatility3\volatility3\framework\deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.registry.hashdump.Hashdump.run) wi
ll be removed in the first release after 2026-09-25. This plugin has been renamed, please call volatility3.plugins.windows.registry.hashdump.Hashdump rather
 than volatility3.plugins.windows.hashdump.Hashdump.
  warnings.warn(

User    rid     lmhash  nthash
E:\Volitility 3\volatility3\volatility3\framework\deprecation.py:105: FutureWarning: This plugin (volatility3.plugins.windows.hashdump.Hashdump) has been re
named and will be removed in the first release after 2026-09-25. Please ensure all method calls to this plugin are replaced with calls to volatility3.plugin
s.windows.registry.hashdump.Hashdump
  warnings.warn(

Administrator   500     aad3b435b51404eeaad3b435b51404ee        31d6cfe0d16ae931b73c59d7e0c089c0
Guest   501     aad3b435b51404eeaad3b435b51404ee        31d6cfe0d16ae931b73c59d7e0c089c0
DefaultAccount  503     aad3b435b51404eeaad3b435b51404ee        31d6cfe0d16ae931b73c59d7e0c089c0
WDAGUtilityAccount      504     aad3b435b51404eeaad3b435b51404ee        69dbee1a98d4f53fbccb1fe5ce37c851
John Doe        1001    aad3b435b51404eeaad3b435b51404ee        ecf53750b76cc9a62057ca85ff4c850e
```

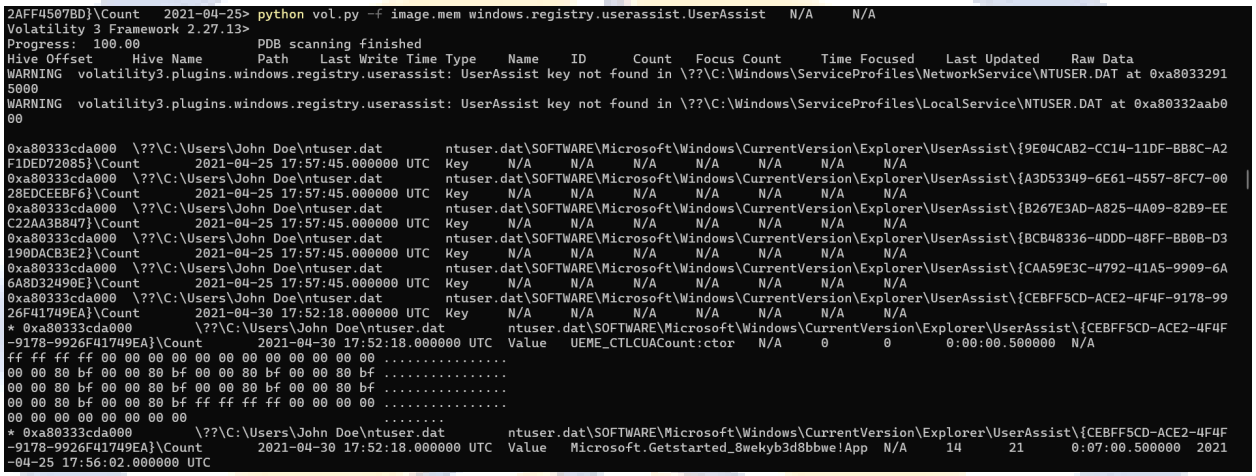
Step 12: User Activity (Registry)

Print out the Windows Registry "UserAssist" key to see what programs were run by the user.

Command:

PowerShell

python vol.py -f [ImageName] windows.registry.userassist.UserAssist



Step 13: Registry Hives

List all available Windows Registry hives found in memory.

Command:

PowerShell

python vol.py -f [ImageName] windows.registry.hivelist.HiveList

```
PS E:\Volitility 3\volatility3> python vol.py -f image.mem windows.registry.hivelist.HiveList
Volatility 3 Framework 2.27.1
Progress:  100.00               PDB scanning finished
Offset  FileFullPath    File output

0xa8032f064000          Disabled
0xa8032f089000  \REGISTRY\MACHINE\SYSTEM        Disabled
0xa8032f135000  \REGISTRY\MACHINE\HARDWARE      Disabled
0xa8033262f000  \Device\HarddiskVolume1\Boot\BCD        Disabled
0xa8032fac1000  \SystemRoot\System32\Config\SOFTWARE    Disabled
0xa8032f632000  \SystemRoot\System32\Config\DEFAULT     Disabled
0xa803327aa000  \SystemRoot\System32\Config\SECURITY    Disabled
0xa8033281e000  \SystemRoot\System32\Config\SAM Disabled
0xa80332915000  \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT        Disabled
0xa80332a95000  \SystemRoot\System32\Config\BBI Disabled
0xa80332aab000  \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT  Disabled
0xa80333cda000  \??\C:\Users\John Doe\ntuser.dat        Disabled
0xa80333d60000  \??\C:\Users\John Doe\AppData\Local\Microsoft\Windows\UsrClass.dat      Disabled
0xa803346e5000  \??\C:\Windows\AppCompat\Programs\Amcache.hve    Disabled
0xa80334a27000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.StartMenuExperienceHost_10.0.19041.610_neutral_neutral_cw5n1h2
txyewy\ActivationStore.dat      Disabled
0xa80334a9e000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\Settings\settings.dat      Disabled
0xa80334b8b000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.Search_1.14.0.19041_neutral_neutral_cw5n1h2txyewy\ActivationSt
ore.dat Disabled
0xa80334cd8000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.Search_cw5n1h2txyewy\Settings\settings.dat       Disabled
0xa80334d62000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.LockApp_10.0.19041.423_neutral__cw5n1h2txyewy\ActivationStore.dat Disa
bled
0xa8033522b000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.LockApp_cw5n1h2txyewy\Settings\settings.dat      Disabled
0xa8033542c000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.YourPhone_1.21022.215.0_x64__8wekyb3d8bbwe\ActivationStore.dat    Disa
bled
0xa80335548000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.YourPhone_8wekyb3d8bbwe\Settings\settings.dat    Disabled
0xa803354da000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.CBS_120.2212.551.0_x64__cw5n1h2txyewy\ActivationStore.da
t       Disabled
0xa80335d1e000  \??\C:\Users\John Doe\AppData\Local\Packages\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\Settings\settings.dat    Disabled
0xa8032f785000  \??\C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\dosvcState.dat Disabled
0xa80336f5b000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.ShellExperienceHost_10.0.19041.610_neutral_neutral_cw5n1h2txye
wy\ActivationStore.dat  Disabled
0xa80336ff2000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.ShellExperienceHost_cw5n1h2txyewy\Settings\settings.dat  Disabled
0xa803326c3000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.WindowsStore_11910.1002.5.0_x64__8wekyb3d8bbwe\ActivationStore.dat   D
isabled
```

Step 14: Dumping Registry Hives

Dump a specific hive (e.g., ntuser.dat) based on a keyword filter to a "dump" folder.

Command:

PowerShell

```
python vol.py -f [ImageName] -o "dump" windows.registry.hivelist --filter Doe\ntuser.dat --dump
```

```
PS E:\Volitility 3\volatility3> python vol.py -f image.mem -o "dump" windows.registry.hivelist --filter Doe\ntuser.dat --dump
Volatility 3 Framework 2.27.1
Progress:  100.00               PDB scanning finished
Offset  FileFullPath    File output

0xa80333cda000  \??\C:\Users\John Doe\ntuser.dat        registry.ntuserdat.0xa80333cda000.hive
```

Step 15: Printing Registry Keys

Print a specific Windows Registry key to inspect its contents.

Command:

PowerShell

```
python vol.py -f [ImageName] windows.registry.printkey --key
```

"Software\Microsoft\Windows\CurrentVersion" --recurse

```
Trust Providers\Software Publishing     State   146432  False
-       0xa80333d60000  Key     \??\C:\Users\John Doe\AppData\Local\Microsoft\Windows\UsrClass.dat\Software\Microsoft\Windows\CurrentVersion    -       -  -
-       0xa803346e5000  Key     \??\C:\Windows\AppCompat\Programs\Amcache.hve\Software\Microsoft\Windows\CurrentVersion -       -       -
-       0xa80334a27000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.StartMenuExperienceHost_10.0.19041.610_neutral
_neutral_cw5n1h2txyewy\ActivationStore.dat\Software\Microsoft\Windows\CurrentVersion    -       -       -
-       0xa80334a9e000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\Settings\settings.dat\S
oftware\Microsoft\Windows\CurrentVersion        -       -       -
-       0xa80334b8b000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.Search_1.14.0.19041_neutral_neutral_cw5n1h2txy
ewy\ActivationStore.dat\Software\Microsoft\Windows\CurrentVersion       -       -       -
-       0xa80334cd8000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.Search_cw5n1h2txyewy\Settings\settings.dat\Software\Microsoft
\Windows\CurrentVersion -       -       -
-       0xa80334d62000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.LockApp_10.0.19041.423_neutral__cw5n1h2txyewy\Activati
onStore.dat\Software\Microsoft\Windows\CurrentVersion    -       -       -
-       0xa8033522b000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.LockApp_cw5n1h2txyewy\Settings\settings.dat\Software\Microsoft\Window
s\CurrentVersion        -       -       -
-       0xa8033542c000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.YourPhone_1.21022.215.0_x64__8wekyb3d8bbwe\ActivationS
tore.dat\Software\Microsoft\Windows\CurrentVersion       -       -       -
-       0xa80335548000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.YourPhone_8wekyb3d8bbwe\Settings\settings.dat\Software\Microsoft\Wind
ows\CurrentVersion      -       -       -
-       0xa803354da000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.CBS_120.2212.551.0_x64__cw5n1h2txyewy\Ac
tivationStore.dat\Software\Microsoft\Windows\CurrentVersion      -       -       -
-       0xa80335d1e000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\Settings\settings.dat\Software\Micros
oft\Windows\CurrentVersion      -       -       -
-       0xa8032f785000  Key     \??\C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\dosvcState.dat\Soft
ware\Microsoft\Windows\CurrentVersion    -       -       -
-       0xa80336f5b000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.ShellExperienceHost_10.0.19041.610_neutral_neu
tral_cw5n1h2txyewy\ActivationStore.dat\Software\Microsoft\Windows\CurrentVersion        -       -       -
-       0xa80336ff2000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.ShellExperienceHost_cw5n1h2txyewy\Settings\settings.dat\Softw
are\Microsoft\Windows\CurrentVersion    -       -       -
-       0xa803326c3000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.WindowsStore_11910.1002.5.0_x64__8wekyb3d8bbwe\Activat
ionStore.dat\Software\Microsoft\Windows\CurrentVersion   -       -       -
-       0xa8033704c000  Key     \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.WindowsStore_8wekyb3d8bbwe\Settings\settings.dat\Software\Microsoft\W
indows\CurrentVersion   -       -       -
-       0xa80337088000  Key     \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\windows.immersivecontrolpanel_10.0.2.1000_neutral_neutral_cw5n1h
2txyewy\ActivationStore.dat\Software\Microsoft\Windows\CurrentVersion    -       -       -
-       0xa80338669000  Key     \SystemRoot\System32\config\DRIVERS\Software\Microsoft\Windows\CurrentVersion   -       -       -
```